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SAKAL ROY

27c1ff43-4607-40aa-81c5-bd...

Course Info: MAT_456_SKY_...
https://shorturl.at/beim5

Add comment

Sakal Roy

Jan 21, 2025

FYI

MAT110_FALL_2024_Secti...
PDF

1 class comment

Sakal Roy • Jan 21, 2025

@everyone

This is already the increased version (maximum around 1.25 marks) from rounding up of marks in individual problems.

Also, I delayed the submission yesterday to ensure that no one is cut off for a grade change for less than 1 marks.

Any further increase is not possible.

If two grade changes are close, there is high probability that the initial grade was lower and by rounding up and partial mark allocation it has somehow been above the grade level.

Hope you understand,

Take care. Best wishes.

Add class comment...

Sakal Roy

Jan 20, 2025

@everyone

Script checking time (mid+final) for MAT 110_Fall 2024_sec=05 and sec_06: 5:00 PM -6:00 PM, 20 Jan, 2025 (MONDAY)

I'll send detailed result analysis and full grade sheet (quiz, assignment marks will be there) also before submitting to USIS at the end of today.

Since, BRACU requests to submit everything within today so I've to submit scripts also. Will Share the detailed result analysis to maintain transparency of assessment and give you a document to reflect even after the script is submitted to BRACU.

I took the time to ensure that you obtain no less grades than what you deserve within the time constraints set by BRACU.

Best wishes.

1 class comment

AHNAF ABTAHI FAHID • Jan 20, 2025

Sir what is the room number?

Add class comment...

Sakal Roy

Jan 12, 2025

A few things to remember:

Stay focused during the exam, write down the steps properly before answering any question.

This will help your think clearly and get rid of silly mistakes.

In exam hall you're only allowed to communicate with your invigilator and only one friend, that friend is not human, its your CALCULATOR.

I hope you'll maintain academic integrity and value that over anything.

Best wishes for you all.

1 class comment

SADIA AFRIN MALIHA • Jan 12, 2025

Thank you sir

Add class comment...

Sakal Roy

Jan 11, 2025

@everyone

FINAL EXAM REVIEW session Recording:

https://classroom.google.com/u/8/c/NzI4Mjg4NTE0NTIw

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https://bdren.zoom.us/rec/share/41k_tXBrOdqhS6WFymj7V_5fB1vOdwspnBndmADCgg3ueZBh8dq2igzmlWNUGQw.XB2xBM1-xgREyiYo

https://bdren.zoom.us/rec/share/41

 Add comment



Sakal Roy
Jan 10, 2025



Sakal Roy is inviting you to a scheduled BdREN Zoom meeting.

Topic: MAT110: FINAL EXAM REVIEW
Time: Jan 11, 2025 07:00 PM Astana, Dhaka

--> Join BdREN Zoom Meeting from Laptop or Mobile:
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdyxs9lbolJOI9.1&omn=96189208345>

Meeting ID: 231 308 1916
Password: savelife

 Add comment



Sakal Roy
Jan 8, 2025



@everyone
TODAYS ONLINE SESSON RESOURCES:
RECORDING:
https://bdren.zoom.us/rec/share/R3UvwOSs2DQglp3XglO6_VeOck33Yo-Bp_gM_6PHOndKFzgTVqUycdPw1T4ENhKo.ZMAAn5HtxMAwhO-YF

BRAINSTORMING BOARD: (Open with draw.io)
https://drive.google.com/file/d/1SnczLpi8x6W1FVrrKsU3tg8n7TH3_CVE/view?usp=sharing

COMMENT HERE WITH YOUR FAVOURABLE DATES AND TIMES FOR FINAL EXAM REIEW CLASS.

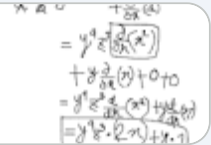
Error - Zoom

<https://bdren.zoom.us/rec/share/R3>



MAT110(LAGRANGE+COIC...

Unknown



 1 class comment



Sakal Roy • Jan 8, 2025

Meeting Summary for MAT110
Jan 08, 2025 06:30 PM Astana, Dhaka ID: 231 308 1916

Quick recap

Sakal led a practice session on partial derivatives, explaining concepts such as the chain rule, Taylor's theorem, and the use of calculators in solving complex equations. He also discussed the application of these concepts in various fields, including mathematics and medicine, and introduced the concept of gradient and its role in solving problems related to partial derivatives. The conversation ended with Sakal emphasizing the importance of algebraic skills, solving equations, and understanding the properties of conic sections for the upcoming exam.

Next steps

Students to focus on application of formulas rather than proofs for the upcoming exam

Students to practice solving Lagrange multiplier problems for optimization

Students to review gradient, divergence, and directional derivative concepts

Students to review formulas for conic sections

Students to familiarize themselves with coordinate system conversions

Students to refer to the official course materials and practice problems provided by the course coordinator

Students to utilize online calculators and solvers for complex calculations during practice Summary

Partial Derivatives Practice Session

Sakal is leading a practice session on partial derivatives. He shares a practice problem and gives 10 minutes for Asadul and others to attempt it. The problem involves finding the partial derivatives of a function with three variables. Sakal explains that they will apply the techniques they have learned, such as the chain rule and Taylor's theorem, to solve the problem. He also mentions that they will practice problems of varying orders of partial derivatives. Asadul and Shithi provide some input during the session.

Partial Differentiation and Chain Rule

Sakal discussed the process of partial differentiation and its application to various functions. He explained the concept of treating variables as constants and the importance of considering the order of differentiation. Sakal also discussed the use of the chain rule and the product rule in differentiation. He demonstrated these concepts using specific functions, including x^2 , y^4 , and z^3 . Sakal also emphasized the importance of simplifying expressions and using calculators to solve complex equations. He concluded by stating that the process of differentiation is crucial in understanding the behavior of functions and their derivatives.

Taylor Series and Lagrange Multipliers

Sakal provides guidance on using calculators carefully, especially for handling brackets and evaluating functions. He suggests practicing online solvers and visualizations for exam preparation. Sakal explains Taylor series approximations for single and multivariable functions, deriving formulas for linear and quadratic approximations. He recommends practicing these approximations with various function examples. Sakal also introduces the concept of using Lagrange multipliers to find maximum and minimum values of a function subject to constraints, providing an example problem to solve.

Exploring Gradient and Vector Operators

Sakal discussed the concept of gradient and its application in various fields, including mathematics and medicine. He explained the idea of a gradient as a slope or rate of change, and how it can be used to find the maximum or minimum of a function. Sakal also discussed the concept of a vector differential operator and its role in solving problems related to partial derivatives. He used examples from mathematics and medicine to illustrate these concepts,



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including the use of gradient in medical imaging and the application of partial derivatives in solving equations. The conversation ended with Sakal asking for ideas on how to further apply these concerns.



techniques. He emphasized the need for algebraic skills and the use of substitution to solve equations. He also mentioned the concept of partial derivatives and the use of multipliers in solving equations. Sakal also touched on the topic of maximum and minimum values of a function subject to certain constraints. He used examples to illustrate these concepts and encouraged the use of visualization tools to better understand the shapes of equations. Asadul asked a question about the maximum and minimum error, to which Sakal responded by explaining the concept of gradient and the use of multipliers. The conversation ended with Sakal suggesting a problem for further discussion.

Exploring Hyperbolas, Ellipses, and Parabolas

Sakal and Parthib discussed various mathematical concepts, including hyperbolas, ellipses, and parabolas. They focused on the properties of these shapes, such as their equations, foci, and asymptotes. Sakal emphasized the importance of understanding the basic ideas and formulas behind these concepts, and suggested practicing with example problems. They also touched on the idea of sketching the graphs of these shapes. However, the conversation ended without a clear resolution or next steps.

Conic Sections and Exam Preparation

Sakal explains the key concepts for the upcoming exam, focusing on conic sections and their equations. He covers parabolas, hyperbolas, and ellipses, providing formulas and methods to convert between different equation forms. Sakal emphasizes practicing problems from the official course materials and following the course coordinator's advice. He assures students that proofs will not be required for the exam, and the focus will be on understanding formulas and their applications. Sakal also briefly mentions polar coordinates but states that solving problems in polar form will not be part of the exam. Overall, he aims to prepare students with the essential topics and problem-solving strategies for the exam, encouraging them to practice and seek additional resources if needed.

AI-generated content may be inaccurate or misleading. Always check for accuracy.

Add class comment...

Sakal Roy

Jan 6, 2025

@everyone

LAST ONLINE SESSION:
LAGRANGE MULTIPLIER+ Coordinate Geometry part will be discussed in this session.
We'll focus on problem solving techniques for these topics. No proof required, only application in the suggested practice problems (find in my Google Doc or from BRACU practice problem drive)
I'll announce another session for FINAL EXAM TOPICS REVIEW before the FINAL EXAM to discuss exam strategies. Hope that helps. Best wishes.

=====

Sakal Roy is inviting you to a scheduled BdREN Zoom meeting.

Topic: MAT110
Time: Jan 8, 2025 06:30 PM Astana, Dhaka

--> Join BdREN Zoom Meeting from Laptop or Mobile:
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdyxs9lbolJOi9.1&omn=96189208345>

Meeting ID: 231 308 1916
Password: savelife

Add comment

Sakal Roy

Jan 5, 2025

@everyone

The main assignment deadline and also late submission deadline with (a 10% penalty per day late, up to three days.) is over now.
It was supposed to be closed by now.
In case any of you missed it due to any unfortunate technical/personal issues, I've requested your ST @Sadia Afrin to keep it open for 6 more hours.
Submit within 10:00 AM, 5 jan,2025 (SUN) (within 6 hours from now) in google form.
Otherwise, the assignment won't even be accepted as late submission deduction.
It'll be evaluated by ST, so there's no point emailing me neither the ST. It has to be in google form as suggested. Please cooperate and if needed check the following to avoid any further technical issues. Best wishes.

=====

- **Refresh the Page**

Press F5 or refresh the browser to reload the form.

- **Clear Browser Cache**

Go to your browser's settings and clear the cache and cookies. Then, reopen the form.

- **Use an Incognito/Private Window**

Open the form in an incognito or private window. This will bypass any stored cookies or extensions that might interfere.

- **Try a Different Browser**

If you're using Chrome, try Firefox, Edge, or Safari to see if the issue persists.

- **Check Internet Connection**

Ensure your internet connection is stable.

- **Disable Extensions**

Browser extensions might block some elements of the form. Temporarily disable them and reload the form.

- **Try on Another Device**

If the issue remains, open the form on another device or smartphone to check if the problem is with your current device.

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Sakal Roy
Dec 31, 2024



Announcement:
I'm cancelling todays schedule consultation hour from **3:00 pm-4:30 pm**.
Will announce a mega consultation hour later.

***This announcement is only for consultation.
Class will go as scheduled.***

Add comment



Sakal Roy
Dec 28, 2024



- FUN FACT 1: What will be the value of $6 \div 2(1+2)$?** Give your opinion.
Then check all the calculating device available to you (i.e. calculator, calculator in mobile, google calculator, programming software etc.)
EXPLORE:<https://youtu.be/vaitsBUyiNQ?si=dPvDid6Ix4VGRBE5>
<https://youtu.be/QOprzEtP19s?si=NwknXUvzhPbWTMsL>
(5 min)
- Gradients and Partial Derivatives**
<https://www.youtube.com/watch?v=GkB4vW16QHI>
(10 min)
- What are derivatives in 3D? Intro to Partial Derivatives**https://www.youtube.com/watch?v=EoEV5-_mLeM
(10 min)
- CalcBLUE 2 : Ch. 2.4 : Derivatives & Rates of Change**https://www.youtube.com/watch?v=DYrs_cOgQE8&list=PL8erLOpXF3JZZTnqjginERYYEi1WpLE_V&index=15
(10 min)
- FUN FACT 2: WHO'LL EAT WHAT?** (See the puzzle in attached image and work out all the possible scenarios)
Believe it or not, Solving this puzzle helps drawing tree diagrams for Chain Rule for Partial Derivatives.
(10 min)
- Chain rule: See selected parts from HOWARD ANTON from Course Info: *GOOGLE DOC*
(15 min)
- SELF-STUDY:
See discussion regarding "differential" from HOWARD ANTON
(See selected parts from Course Info: *GOOGLE DOC*)

(10 min read)

FUN FACT:

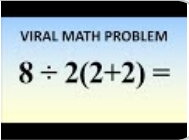
Believe it or not, Solving this puzzle helps drawing tree diagrams for Chain Rule for Partial Derivatives.



FOOD PUZZLE.png

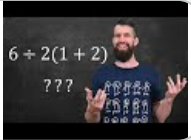
$8 \div 2(2+2) = ?$ Mathematicia...

YouTube video • 5 minutes



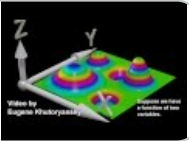
Math Prof answers $6 \div 2(1+2)$...

YouTube video • 6 minutes



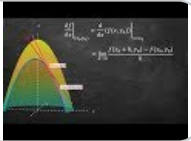
Gradients and Partial Deriv...

YouTube video • 5 minutes



What are derivatives in 3D?...

YouTube video • 8 minutes



CalcBLUE 2 : Ch. 2.4 : Deriv...

YouTube video • 4 minutes



Add comment



Sakal Roy
Dec 28, 2024



@everyone
I'm on my way to scheduled consultation. I will be a bit late.
Will Start at 12:00 pm (instead of 11:30 am) and consultation will be extended upto 1:30 pm (instead of 1:00 pm).
If you're already there please wait.
Sorry for the inconvenience and thanks for understanding.

Add comment



Sakal Roy
Dec 27, 2024 (Edited Dec 27, 2024)



@ everyone.
Hello,
Hope you are all doing fine. As you all know BRACU has suggested an online make up class on 27 December, 2024 (FRIDAY). (today).
However, since we've class on SATURDAY 28 December, 2024 (tomorrow).
So, I won't take any separate online session today.

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Rather, I'm going to request you to explore a few FUN FACTS by yourself which will make us ready in our tomorrow's session:

- FUN FACT 1: What will be the value of $\frac{d}{dx} \ln(x^2 + 1)$? Give your opinion.

- FUN FACT 2: WHO'LL EAT WHAT?(See the puzzle in attached image and work out all the possible scenarios)
Believe it or not, Solving this puzzle helps drawing tree diagrams for Chain Rule for Partial Derivatives

DON'T SEE ANY RESOURCES, JUST ANSWER WHAT YOUR MIND SAYS.SEE YOU ALL IN TOMORROW'S CLASS.
BEST WISHES.

FUN FACT:

Believe it or not, Solving this puzzle helps drawing tree diagrams for Chain Rule for Partial Derivatives.



FOOD PUZZLE.png

Add comment



Sakal Roy
Dec 21, 2024



@everyone
I'm on my way to scheduled consultation. I will be a bit late.
Will Start at 12:00 pm (instead of 11:30 am) and consultation will be extended upto 1:30 pm (instead of 1:00 pm).
If you're already there please wait.
Sorry for the inconvenience and thanks for understanding.

Add comment



Sakal Roy
Dec 19, 2024



@everyone
Practice these:(google doc contains this one.)
There's more following the same principle which are suggested by BRACU.
Problems from howard anton and online solver with solved examples are also available in google doc. But try all of this first. We'll discuss the problems you face difficulties in the next class and move to the next topics afterward.

<https://drive.google.com/file/d/1nU2rOmvm6aPQunoaAyBrKrOEkgtkGOoi/view?usp=drivesdk>

Add comment



Sakal Roy
Dec 19, 2024



Will start todays class at 3:50 pm.
First 20 minutes (3:30 p

- m3:50 pm), I would like to ask everyone to try the "Hole in the hand" illusion yourself. Link is given in last announcement.
And, obviously, try to find out why you think you can see a hole in your hand. Take time.
I'll start the class around 3:50 pm listening to your experiences and then diving into the world of 3D.
I'll try to show abd discuss the given links above.
You can read it beforehand since today we'll try to share all our thoughts regarding seeing 3D, feeling 3D, 3D coordnate, vector and all thats necessary.
Best wishes.

Add comment



Sakal Roy
Dec 18, 2024



ASSIGNMENT:
<https://docs.google.com/document/d/1eoQkTSSsw3Pf9LADGulcVLl6CeZQYUtVnKTjbpxhgM/edit?usp=sharing>
Also, attaching the pdf version.
BEST WISHES.

MAT110_SKY_FALL-2024_A...
Google Docs



MAT110_SKY_FALL-2024_A...
PDF



Add comment



Sakal Roy
Dec 16, 2024



@everyone
Announcement:
You might've already know that there will be no SUN-TUES class on 17 december, 2024 (TUES) class as per BRACU academicc calendar.

BRACU swapped it with Mon-Wed class.

But MY scheduled Consultation hour on 17 december, 2024 (TUES) will go on as usual.

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Sakal Roy

Dec 14, 2024



@everyone

TOPICS AND RESOURCES FOR TODAYS AND NEXT CLASS.

1. SINGLE VARIABLE TO MULTIVARIABLE CALCULUS:<https://youtu.be/gsUgDpGWk-M?si=FXDU8OKTWkWIpjCR>

2. Taylor and maclaurin polynomial and series:

Brief idea about interpreting single variable function case to multivariable function

3. Maxima, minima: critical point and stationary points, concave up and down and inflection point:

Brief idea about interpreting single variable function case to multivariable function

4. LIFE AND MATH in 3D:

>>TRY THIS AT HOME:

Hole in the hand:

<https://www.instructables.com/Hole-in-Your-Hand-Illusion/>

>>EXPLORE 3D coordinate visualization:

Mission impossible:

<https://www.youtube.com/watch?v=qtA0JS1BaY&t=1s>

Harry potter:

<https://www.youtube.com/watch?v=uT9RdFL4SaE>

5. TO BE DISCUSSED IN NEXT CLASS:

>>A little bit of recap from single variable and introduction to multivariable graph:

https://www.youtube.com/watch?v=e2TJpv39SJo&list=PL8erLQpXF3JZZTnqjginERYYEi1WpLE_V&index=4

<https://www.youtube.com/watch?v=vjUy9kevd3Y&list=PL8erLQpXF3JYm7VaTdKDaWc8Q3FuP8Sa7&index=15>

https://www.youtube.com/watch?v=gbjOYfdovxU&list=PL8erLQpXF3JZZTnqjginERYYEi1WpLE_V&index=5

https://www.youtube.com/watch?v=_sjTl6q5hHc&list=PL8erLQpXF3JZZTnqjginERYYEi1WpLE_V&index=7

https://www.youtube.com/watch?v=_sjTl6q5hHc&list=PL8erLQpXF3JZZTnqjginERYYEi1WpLE_V&index=8

https://www.youtube.com/watch?v=CcdUud9BFck&list=PL8erLQpXF3JZZTnqjginERYYEi1WpLE_V&index=8

<https://youtu.be/GkB4vW16QHI>

https://youtu.be/EoEV5-_mLeM

>> Explore Cartesian product using playing card example

https://en.wikipedia.org/wiki/Cartesian_product

>>Explore 2D and 3D geometry and vector basics

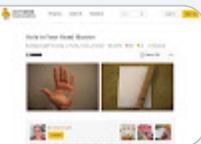
<https://www.geogebra.org/graphing?lang=en>

<https://www.geogebra.org/3d?lang=en>

<https://youtube.com/watch?v=Ej3ZVxlJjfo&feature=share9>

Hole in Your Hand Illusion : ...

<https://www.instructables.com/Hole-in-Your-Hand-Illusion/>



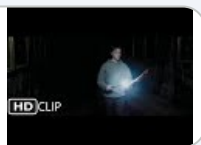
Mission Impossible: Ghost ...

YouTube video • 7 minutes



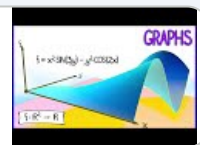
Pettigrew on The Marauder'...

YouTube video • 4 minutes



CalcBLUE 2 : Ch. 1.1 : Graph...

YouTube video • 1 minute



CalcBLUE 1 : Ch. 3.1 : Coord...

YouTube video • 2 minutes



Cartesian product - Wikiped...

[https://en.wikipedia.org/wiki/Cartes](https://en.wikipedia.org/wiki/Cartesian_product)



Graphing Calculator - Geo...

<https://www.geogebra.org/graphing>



3D Calculator - GeoGebra

[https://www.geogebra.org/3d?lang](https://www.geogebra.org/3d?lang=en)



Everything You Need to Kn...

YouTube video • 17 minutes



CalcBLUE 2 : Ch. 1.2 : Visual...

YouTube video • 2 minutes



CalcBLUE 2 : Ch. 1.4 : Multiv...

YouTube video • 4 minutes



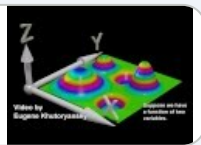
CalcBLUE 2 : Ch. 1.5 : Towar...

YouTube video • 0 minutes



Gradients and Partial Deriv...

YouTube video • 5 minutes



What are derivatives in 3D?...

YouTube video • 8 minutes



What are the big ideas of M...

YouTube video • 16 minutes



Add comment



Sakal Roy

Dec 12, 2024



@everyone

We'll start class at 3:45 PM.

From 3:30 PM -3:45 PM these 15 minutes are dedicated for your thinking the2 questions: (1 (a) and 1(b). Let me know what you vote for in comments or in the beginning of class.

The rest (2,3,4) we'll discuss in class.

See you all in class.

1. Circular reasoning :

Can we apply Lahopitas rule in determining the following limits:

a) $\lim_{x \rightarrow 0} [\sin(x) / x] = ?$ [Yes/No/Maybe]

b) $\lim_{x \rightarrow 0} [(1+x)^{1/x}] = ?$ [Yes/No/Maybe]

2. Taylor and maclaurin polynomial and series:

Brief idea about interpreting single variable function case to multivariable function

3. Maxima, minima: critical point and stationary points, concave up and down and inflection point:

Brief idea about interpreting single variable function case to multivariable function

4. LIFE AND MATH in 3D:

>>Mission impossible movie clip + Harry potter goblet of fire: first task

>>Focus on 3D coordinate visualization,

>>Watch videos from google doc regarding multivariable function

>> Explore Cartesian product using playing card example

>>Explore 3D geometry and vector basics



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MAT 110(Fall-25)20

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Archived classes

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MAT110_FALL2024_SKY 6

Transformation of curvilinear coordinates (rectangular, cylindrical, spherical coordinates)

Regards,
Sanjeeda Nazneen

Stream

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People

=====

Add comment

Sakal Roy
Dec 1, 2024

Hello, everyone, hope you are all well.
I have 7 announcements for you.
Look into them one by one.

1) In the first 2 lectures we'll recap MID-TERM topics.

It is just to make sure that we can grasp the FINAL TOPICS related to multivariable calculus, without any doubt, confusion or hesitation.

2) Among these first 2 lectures,
I'll allocate first class:

Thursday, 5 dec, 024 for your self study.

That means no offline or online class will be held from me on this THURSDAY.

3) Also, there'll be no consultation hour from me on this week at TUESDAY, 3 dec, 2024.

To make this up, apart from regular consultations, I'll announce another mega consultation hour later in next weeks.

4) That means, I'll meet you all in person at class a after MID on next SATURDAY, 7 december, 2024.

5) I'll announce an online assignment later in next week fro MID topics only to ensuure that we're all set to continue FINAL TOPICS.

You'll have about 2 weeks time to submit those online assignments just from the MID TERM TOPICs.

6) Meanwhile, recap MID-TOPICs from google doc, see the 2 recorded lectures before MID EXAM, which are shared with you.

Feel free to let me know your query confusion in email or here.
I'll start next class on SAT focusing on those specific problems or topics.
You can have open discussion here also if you wish.

7) After 2 lectures, we'll continue our journey into the mutivariable calculus targetting FINAL EXAM. So, in case anyone is wondering why we would learn multivariable calculus watch the attached video that I promised to show in the beginning of the course.

<https://www.youtube.com/watch?v=qtAQJS1BaY&t=1s>

We'll see it again in next classes together.
Let me know what do you think about this video.
Enjoy!

Mission Impossible: Ghost...
YouTube video • 7 minutes

Add comment

Sakal Roy
Nov 22, 2024

Hello everyone,
See the recording of review class here:
https://bdren.zoom.us/rec/share/Y1WlkefybBAOn89Ua5qzFKhtSP4f9qbzIGYcV5fJy9VrHVKgdx7deSC9yJDVI_G.Ai6EwPSLVxp5dbiY
See the writeup if needed here:
Open with Draw.io:
<https://drive.google.com/file/d/17MVXC96yB-mt2N5dUdW5JziuHVAWYTQK/view?usp=sharing>
Try to avoid calculation mistakes, use calculator properly.
Focus on the basic idea and steps on each topics to avail partial marks even if there's slight mistake
Best wishes.

Error - Zoom

<https://bdren.zoom.us/rec/share/Y1>

MAT110_FALL 2024_MID TE...
Unknown

Add comment

Sakal Roy
Nov 22, 2024

Hello, everything is nearly sorted in my end.
I'll start at 7:30 pm.
Join in the same link.
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdyxs9IbolJOI9.1&omn=91309508372>

Passcode:
savelife

Sorry for the inconvenience.

Add comment

Sakal Roy
Nov 22, 2024

https://classroom.google.com/u/8/c/NzI4Mjg4NTE0NTIw

10/14

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Settings

Hello everyone, due to power outage and internet disruption in my area I can't start at 6:00 pm.
I wish start around 7: 00 pm when everything stabilizes.
You may watch the last class video recording if you haven't for Maxima Minima and Taylor-Maclaurin Recap, I'll



Stream Classwork People

<https://www.youtube.com/watch?v=upywEwTKd8&index=13&t=5s>

Because only these type problems are a bit different.
I'll not discuss any theory in this review class. It'll be based on your comments on any topic or any specific problem or confusion you faced.
So, you can email me or mention is this comment about any topics or problem no beforehand.
I'll try to directly touch those parts only.

2.12 The Squeeze Theorem
YouTube video • 5 minutes



Add comment



Sakal Roy
Nov 21, 2024



WATCH/REWATCH this one.
https://bdren.zoom.us/rec/share/JiQyfg_WgScdbZpINZqxMrbwXR-rY30Tgc72AOPLm29yEsC7bKsprbuiMPUQ8PCh.bndySxqhzjYQ7Hw3

Let me know if we can have a REVIEW SESSION from 6:00 pm tomorrow, FRIDAY, 22 NOVEMBER, 2024.
I'll share joining link if most students are okay with the time. Comment YES/ suggested time here.

- Review class will be recorded and shared with all.
- So, you can share your query here or in email beforehand. Or you can ask yourself joining the review session.
- Make sure your question reaches me , I'll try to explain or solve accordingly, you can even watch recording even if you can't manage to attend the review session.

Best wishes.

Error - Zoom

<https://bdren.zoom.us/rec/share/JiC>



2 class comments



Raiyan Muhtaseem • Nov 21, 2024
Yes

5:30 PM is too early
6:00 PM is perfect for me



Add class comment...



Sakal Roy
Nov 21, 2024



To be discussed in todays class:
1) DIFFERENTIABILITY DISCUSS USING LIMIT DEFINITION

- <https://www.youtube.com/watch?v=53R2f5PLf7k>
- <https://www.mathsisfun.com/calculus/slope-function-point.html>
- <https://www.mathsisfun.com/calculus/derivatives-introduction.html>
- https://www.youtube.com/watch?v=7vhux5TLRmQ&list=PLlwePzQY_wW8qiZD6XYqCnibdY37ygbx7&index=1
- https://www.youtube.com/watch?v=eNcg9cKzV1Q&list=PLlwePzQY_wW8qiZD6XYqCnibdY37ygbx7&index=2
- <https://www.mathsisfun.com/calculus/differentiable.html>

- a) The mighty substitution: $x-a = h \Rightarrow x= a+h$,
b) Pros and Cons of LaHopital law:
#Home Task: See how to handle all types of INDETERMINATE FORM in limit from GOOGLE DOC.
#Mathematical Debate:
Can we useLaHopital law in determining the following limit?
a) $\lim (x \rightarrow 0) [\sin (x) / x] = 1$:See GOOGLE DOC
b) $\lim (x \rightarrow \inf) [(1+1/x)^x] = \lim (z \rightarrow 0) [(1+z)^{(1/z)}] = 1$: See GOOGLE DOC
#GOAL: Beware of the TRAP of CIRCULAR REASONING While using Lahopital law.

2) TOPICS OF DIFFERENTIATION :

- <https://www.mathsisfun.com/calculus/derivatives-rules.html>
- <https://www.mathsisfun.com/calculus/implicit-differentiation.html>

- a) Practice suggestion from COURSE INFO GOOGLE DOC, just like H.S.C/GRADE-11-12.
b) What does differential coefficient mean?

3) SUCCESSIVE DIFFERENTIATION : See all practice problem solution from GOOGLE DOC.

- <https://www.mathsisfun.com/calculus/second-derivative.html>
- https://www.youtube.com/watch?v=7su5mypmgdw&list=PLlwePzQY_wW8qiZD6XYqCnibdY37ygbx7&index=8

4) MAXIMA, MINIMA:
RECAP: INCREASING , DECREASING, CONCAVE UP, DOWN, CRITICAL POINT AND STATIONARY POINT from Google DOC
5) TAYLOR AND MACLAURIN POLYNOMIAL AND SERIES See from GOOGLE DOC
a) How to remember or recreate the formula?
=> Detailed PROOF is the best way but just looking for a PATTERN is enough.
b) How to apply it properly?
=> Difference between $f'(x)$ and $f'(a)$
c) What's the difference between polynomial and series?
=> THE TALE of 3 DOTS.

Add comment



Sakal Roy
Nov 20, 2024



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CONSULTATION/CLASS JOINING LINK, same link for both join according to the scheduled time:
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdyxs9lbolJOl9.1&omn=91309508372>
Passcode:



Stream Classwork People

2 class comments

Sakal Roy • Nov 20, 2024
Yeah, since everything is shifted to online tomorrow, no need to take risk. Stay safe.

Add class comment...

Sakal Roy
Nov 20, 2024

URGENT:
Announcement from BRACU, MNS.
I'll share joining links from class/consultation for tomorrow here. Stay safe.
=====

Dear Respected Faculties,

Hope you are doing well. I am writing you to let you know that there is a central decision regarding tomorrow's class. You are instructed to conduct tomorrow's class online.

Please make the necessary arrangements to conduct tomorrow's class online.

Thank you very much.

Sincerely,
Rafsanjany Jim
=====

Add comment

Sakal Roy
Nov 20, 2024

MEGA CONSULTATION HOUR:
To aid mid term preparation,
I'm announcing **an extra consultation hour on tomorrow from 11:00 AM to 2:00 PM.**
You can meet me and clear your query/confusions, also email me in my official email in case you can't meet me in person in the in consultation hour. I'll try to assist you as much as possible in time.
ONLINE REVIEW SESSION:
I also wish to take an extra Online Review session focusing on exam strategy.
I think **22th november after 5:00 pm** is preferable to avoid clash with a anyones exam time.
However, give your opinion in comments here regarding when itsbconvenient to start the review session after 5:00 pm. I'll post online links accordingly.

3 class comments

Sakal Roy • Nov 20, 2024
Bracu, MNS just announced online class for tomorrow probably observing severe unrest thats going on. I'm making another announcement mentioning that. Will provide online joining link for class/consultation.

Add class comment...

Sakal Roy
Nov 16, 2024

See the following for a recap of todays lecture.
More details and solved examples are in the google doc.

2.12 The Squeeze Theorem

YouTube video • 5 minutes



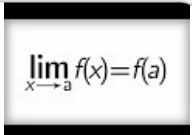
2.14 The definition of contin...

YouTube video • 7 minutes



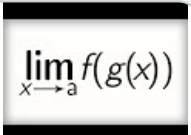
2.15 The main continuity th...

YouTube video • 4 minutes



2.16 Limits and composition...

YouTube video • 7 minutes



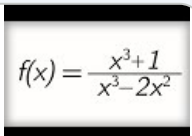
2.17 Discontinuities (and ho...

YouTube video • 3 minutes



6.16 Vertical and horizontal ...

YouTube video • 6 minutes



2 class comments

Sakal Roy • Nov 16, 2024
Yeah, usually you've to answer 5 question.
However, the number of altenative (5 out of 6 or 5 out of more) is decided by course coordinator based on discussion with all faculties.
My suggestion would be to prepare focusing on the basic steps of each topic included ib mid for fall 2024. I'll discuss more in the next class and will also take an extra online review session.

Add class comment...

SADIA AFRIN
Nov 16, 2024 (Edited Nov 16, 2024)

Dear Students,

I hope this message finds you well. Below are my consultation hours for your convenience:

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Saturday:

9:30 AM - 10:50 AM (06D-03C)

12:30 PM - 1:50 PM (06D-02C)

8:00 AM - 9:20 AM (06D-02C)

12:30 PM - 1:50 PM (06D-02C)

Email: sadia.afrin9@g.bracu.ac.bd

★ Kindly email me beforehand with your problem.

★ There will be no consultation during the midterm and final exam weeks.

★ If these times are not convenient, you can visit any other available student tutors for assistance.

Best regards,

Sadia Afrin

Student Tutor

Add comment

Sakal Roy

Nov 11, 2024

@everyone

MSG from Course acoordinator MAT110 FALL 2024.

=====

Dear colleagues,

We would exclude the topics **(Leibnitz's Theorem, Roll's and Mean value theorem)** from the midterm syllabus for the Semester Fall 2024.

Updated Midterm syllabus:

Limits and Continuity

Differentiability and Techniques of Differentiation

Successive Differentiation

Maxima, Minima

Taylor and Maclaurin polynomials and Series for single variable functions

Regards,

Sanjeeda Nazneen

Senior Lecturer

Department of Mathematics and Natural Sciences

BRAC University

=====

Add comment

Sakal Roy

Nov 9, 2024

@everyone

I'll start today's scheduled consultation hour 30 minutes late (from 12:00 pm -1:30 pm) due to unavoidable traffic situations. If anyone already reached BRACU for consultation from me, please wait. Sorry for the inconveniences.

Add comment

Sakal Roy

Nov 8, 2024 (Edited Nov 8, 2024)

Go through the following resources to get a recap of the last classes.you can find these resources in course info google doc also. I'm just posting here for convenience.

In next class we'll discuss special techniques for calculating limits and mathematical way of defining of continuity. Feel free to explore GEOGEBRA to learn plotting piecewise function. It'll help you visualize and comment on the limit for any function instantly.

We'll have a QUIZ on limit ,continuity and Topics of differentiation in the week after next week.

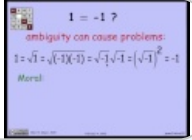
Don't worry about the quiz date. This quiz will be online, no lecture timw will be needed to be occupied for that.

Also, if we can't find any convenient time in last week, it can also be scheduled after mid if needed.

But first: **make sure that we're not forgetting any basic principles while calculating limits. Come to consultation or email me with specific query/confusion and your opinion regarding course topic/assessment style etc if needed.**

1.1.3 Intro to Proofs: Part 2

YouTube video • 7 minutes



Absolute Value in Algebra

https://www.mathsisfun.com/algebi



Limits to Infinity

https://www.mathsisfun.com/calcul



Graphing Calculator - Geo...

https://www.geogebra.org/graphin



Squares and Square Roots i...

https://www.mathsisfun.com/algebi



Limits (An Introduction)

https://www.mathsisfun.com/calcul



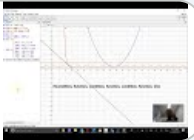
What's Infinity?

https://www.mathsisfun.com/numb



Using Geogebra to create a...

YouTube video • 4 minutes



2 class comments

Sakal Roy • Nov 8, 2024

Don't worry about the date. This quiz will be online, no lecture timw will be needed to be occupied for that. Also, if we can't find any convenient time in last week, it can also be scheduled after mid if needed.

Add class comment...

Sakal Roy

Nov 6, 2024

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MAT110_FALL2024_SKY 6
Dear Students,
Let's welcome SADIA AFRIN,
Student Tutor (ST):



Requesting SADIA AFRIN to let you know further ST consultation info here in google classroom.
I'll update in google doc after that, it'll be kept yellow till then to avoid confusion.

Welcome here, SADIA
Hoping for a nice experience with you, the students and MAT110 in fall 2024.

Add comment

Sakal Roy
Nov 5, 2024 (Edited Nov 5, 2024)

I'll start consultation hour 30 minutes late today (from 3:30 pm-5:00 pm) due to unavoidable traffic situations. If anyone already reached BRACU for consultation from me, please wait. Sorry for the inconveniences.

Add comment

Sakal Roy
Nov 5, 2024

URGENT
Msg from Course Coordinator regarding MID TERM TOPIC ADJUSTMENT based on this semester: FALL 2024's schedule.

- =====
- Limits and Continuity
 - Differentiability and Techniques of Differentiation
 - Successive Differentiation, (Leibnitz' product rule is excluded for Fall 2024)
 - Maxima and Minima (Optimization is excluded for Fall 2024), Roll's and Mean value theorem
 - Taylor and Maclaurin polynomials and Series for single variable functions

=====

I'm also attaching here updated practice sheet provided by Course Coordinator where which parts are excluded in this semester are labeled as [Excluded] or [Excluded for FALL 2024] in the practice sheets and crossed out in the practice problem list from books.

The rest of the topics and practice problems remains same.
For convenience I've already highlighted such topics or practice problems in the course info google doc as **YELLOW**.Hope that helps.
See you all in next class.

[Practice Problems from bo...](#)
PDF



[Rolle's and Mean value The...](#)
PDF



[Successive differentiation ...](#)
PDF



Add comment

Sakal Roy
Nov 3, 2024

Hello everyone,
Here's this semesters course info : all in one place. Hope that helps.
Best wishes..

MAT110_FALL2024 SKY
To share this document to anyon

- **eUSE the following link:**
<https://shorturl.at/w6zWV>
or
SCAN the following QR Code:

*You can share this link with the following foreword to anyone or
Keep it confidential before the end of this semester.*



Add comment